

Numerical Study of Two-Dimensional Quantum Double-Slit Interference using a Crank–Nicolson Scheme

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This report presents a numerical study of the time-dependent Schrödinger equation in two spatial dimensions for a single, non-relativistic particle confined to a finite box. The equation is discretised using a Crank–Nicolson scheme on a uniform grid, leading to a sparse complex linear system at each time step. The associated matrices are assembled in a five-point stencil form and solved with a sparse direct solver from the Armadillo C++ library. As a first validation, the free evolution of a Gaussian wave packet is simulated and the total probability is shown to be conserved to within machine precision throughout the run. Introducing a high barrier with one, two, and three slit openings, we study how the wave packet diffracts and interferes as it traverses the potential. The time evolution of the probability density and of the real and imaginary parts of the wave function are visualised, revealing the build-up of a characteristic interference pattern behind the barrier. Finally, detection probabilities on a distant screen are extracted and compared for the single-, double-, and triple-slit configurations, illustrating quantitatively how the spatial structure of the potential shapes the observed interference fringes.

I. INTRODUCTION

Motivation and Objectives

The aim of this project is to solve numerically the two-dimensional time-dependent Schrödinger equation for a single, non-relativistic particle confined to a finite box and subject to various external potentials. In particular, we will focus on a double-slit setup inside the box, allowing us to reproduce and explore interference patterns analogous to the standard double-slit experiment.

Beyond reproducing a textbook quantum-mechanical phenomenon, the project is intended as a training ground for numerical methods applied to partial differential equations (PDEs). The key methodological aspects include: formulating the Schrödinger equation in position space, applying the Crank–Nicolson method in $2 + 1$ dimensions, working consistently with complex-valued quantities in code, handling indices in a two-dimensional grid, and using conserved quantities such as the total probability.

The main objectives of the report are therefore (i) to implement a stable and accurate Crank–Nicolson solver for the two-dimensional Schrödinger equation, (ii) to use this solver to simulate the time evolution of a Gaussian wavepacket as it propagates through the double-slit geometry, and (iii) to analyse the resulting probability distributions in order to quantify diffraction and interference effects.

The remainder of this article is organised as follows. In Sec. II we introduce the physical and mathematical background, define the computational domain and boundary conditions, and derive the Crank–Nicolson discretisation together with its sparse matrix formulation. Sec. III presents numerical results: we first verify probability conservation for a free wave packet, then analyse the time evolution of the wave function in a double-slit potential, and finally extract detection probabilities for different slit

configurations on a distant screen. Sec. IV summarises the main findings of the project and outlines possible extensions of the present implementation.

All code developed for this report is available via this GitHub repository ¹.

Physical and Mathematical Background

In non-relativistic quantum mechanics, the state of a particle is represented by a vector $|\Psi\rangle$ in a complex Hilbert space. When we work in position space, this state is expressed through the wave function $\Psi(x, y, t)$, which gives the amplitude for finding the particle at position (x, y) at time t . The dynamics of Ψ are governed by the time-dependent Schrödinger equation

$$i\hbar \frac{\partial}{\partial t} \Psi(x, y, t) = \hat{H} \Psi(x, y, t), \quad (1)$$

where $\hat{H}$ is the Hamiltonian operator. For a single particle of mass m in two spatial dimensions and in the presence of a potential $V(x, y)$, the Hamiltonian takes the familiar form

$$\hat{H} = -\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) + V(x, y). \quad (2)$$

It is convenient for numerical work to introduce dimensionless variables by scaling space, time, and energy. After this rescaling, the Schrödinger equation can be written in the simplified form [1]

$$i \frac{\partial u}{\partial t} = -\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} + v(x, y) u, \quad (3)$$

¹ <https://github.uio.no/teresm/Comp.Phys.Project/tree/master/P5>

where $u(x, y, t)$ is the dimensionless wave function and $v(x, y)$ is the corresponding dimensionless potential. In this formulation all variables are dimensionless, which removes the need to track physical units explicitly in the numerical implementation.

The connection between the wave function and observable quantities is provided by the Born rule. In position space, the probability density for detecting the particle at (x, y) at time t is given by

$$p(x, y; t) = |u(x, y, t)|^2 = u^*(x, y, t) u(x, y, t), \quad (4)$$

assuming that $u(x, y, t)$ has been properly normalised. Throughout this project we will analyse the time evolution of this probability density to study how diffraction and interference patterns emerge from the underlying wave dynamics.

Domain, Notation and Boundary Conditions

We consider a square spatial domain $[0, 1] \times [0, 1]$ with time $t \in [0, T]$. The spatial coordinates are discretised using an equal step size h in both directions,

$$x \rightarrow x_i = ih, \quad y \rightarrow y_j = jh, \quad i, j = 0, 1, \dots, M-1,$$

and time is discretised as

$$t \rightarrow t_n = n\Delta t, \quad n = 0, 1, \dots, N_t - 1.$$

The wave function and potential on the grid are denoted by

$$u(x, y, t) \rightarrow u_{ij}^n, \quad v(x, y) \rightarrow v_{ij},$$

so that the numerical solution at time level n can be viewed as a matrix U^n with entries u_{ij}^n .

To model a particle confined inside a box, we impose Dirichlet boundary conditions on the edges of the domain,

$$u(0, y, t) = 0, \quad u(1, y, t) = 0, \quad (5)$$

$$u(x, 0, t) = 0, \quad u(x, 1, t) = 0, \quad (6)$$

for all relevant times t . Physically, these conditions represent infinitely high potential walls surrounding the system. The double-slit geometry and any other configurations studied in this report are encoded in the potential $v(x, y)$ inside the domain. As an initial condition $u(x, y, 0)$, we will typically employ a quantum-mechanical Gaussian wavepacket located inside the box, which then evolves according to Eq. (3).

Tools

All simulations were implemented in modern C++ using the Armadillo linear algebra library [2]. In particular, Armadillo's complex sparse matrix type and the

`spsolve` interface to the SuperLU backend were used to assemble and solve the Crank–Nicolson linear systems efficiently at each time step. Post-processing and visualisation of the data—including plots of the probability density, the real and imaginary parts of the wave function, screen detection probabilities, and the probability-evolution animation—were carried out in Python 3 using NumPy and Matplotlib [3]. This tool chain makes it straightforward to couple a performant C++ time integrator to a flexible, high-level analysis and plotting environment.

In addition, a large language model-based AI assistant (ChatGPT, provided by OpenAI) was used throughout the project as a support tool. The AI was employed to refine the structure and wording of the report, to help translate some explanations into clear scientific English, and to suggest LaTeX formulations of equations and figures based on already derived results. It also assisted in designing and reviewing C++ and Python code snippets (for example, for sparse matrix assembly, time stepping and plotting scripts), which were then tested, validated and, when necessary, modified by the author. The scientific choices, parameter selections and interpretation of the numerical results remain the responsibility of the author.

II. METHODS

Crank–Nicolson discretisation

We start from the dimensionless time-dependent Schrödinger equation, Eq. (3) in the previous section. Rewriting it as an explicit evolution equation for u ,

$$\frac{\partial u}{\partial t} = i \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} - v(x, y) u \right) \equiv F[u(x, y, t)]. \quad (7)$$

For a general equation $\partial u / \partial t = F[u]$, the θ -scheme discretises the time derivative according to

$$\frac{u_{ij}^{n+1} - u_{ij}^n}{\Delta t} = \theta F_{ij}^{n+1} + (1 - \theta) F_{ij}^n, \quad (8)$$

where $F_{ij}^n \equiv F[u_{ij}^n]$. Choosing $\theta = 0$ gives the forward (explicit) Euler method, $\theta = 1$ yields the backward (implicit) Euler method, and $\theta = \frac{1}{2}$ corresponds to the Crank–Nicolson scheme [4]:

$$\frac{u_{ij}^{n+1} - u_{ij}^n}{\Delta t} = \frac{1}{2} (F_{ij}^{n+1} + F_{ij}^n). \quad (9)$$

We discretise the spatial domain as in the previous section: $x_i = ih$, $y_j = jh$ with $i, j = 0, \dots, M-1$, and $t_n = n\Delta t$. The second derivatives in Eq. (7) are approx-

imated by central finite differences,

$$\left. \frac{\partial^2 u}{\partial x^2} \right|_{ij}^n \approx \frac{u_{i+1,j}^n - 2u_{ij}^n + u_{i-1,j}^n}{h^2}, \quad (10)$$

$$\left. \frac{\partial^2 u}{\partial y^2} \right|_{ij}^n \approx \frac{u_{i,j+1}^n - 2u_{ij}^n + u_{i,j-1}^n}{h^2}. \quad (11)$$

Using this notation, the discrete version of F at time level n reads

$$F_{ij}^n = i \left[\frac{u_{i+1,j}^n - 2u_{ij}^n + u_{i-1,j}^n}{h^2} + \frac{u_{i,j+1}^n - 2u_{ij}^n + u_{i,j-1}^n}{h^2} - v_{ij} u_{ij}^n \right], \quad (12)$$

with an analogous expression for F_{ij}^{n+1} .

Inserting these expressions into the general CN formula Eq. (9) and multiplying both sides by Δt gives

$$u_{ij}^{n+1} - u_{ij}^n =$$

$$\begin{aligned} & \frac{i\Delta t}{2} \left[\frac{u_{i+1,j}^{n+1} - 2u_{ij}^{n+1} + u_{i-1,j}^{n+1}}{h^2} + \frac{u_{i,j+1}^{n+1} - 2u_{ij}^{n+1} + u_{i,j-1}^{n+1}}{h^2} - v_{ij} u_{ij}^{n+1} \right] \\ & + \frac{i\Delta t}{2} \left[\frac{u_{i+1,j}^n - 2u_{ij}^n + u_{i-1,j}^n}{h^2} + \frac{u_{i,j+1}^n - 2u_{ij}^n + u_{i,j-1}^n}{h^2} - v_{ij} u_{ij}^n \right]. \end{aligned}$$

It is convenient to introduce the dimensionless parameter

$$r \equiv \frac{i\Delta t}{2h^2}. \quad (13)$$

With this definition, the previous expression can be written compactly as

$$u_{ij}^{n+1} - u_{ij}^n =$$

$$\begin{aligned} & r \left[u_{i+1,j}^{n+1} - 2u_{ij}^{n+1} + u_{i-1,j}^{n+1} + u_{i,j+1}^{n+1} - 2u_{ij}^{n+1} + u_{i,j-1}^{n+1} \right] - \frac{i\Delta t}{2} v_{ij} u_{ij}^{n+1} \\ & + r \left[u_{i+1,j}^n - 2u_{ij}^n + u_{i-1,j}^n + u_{i,j+1}^n - 2u_{ij}^n + u_{i,j-1}^n \right] - \frac{i\Delta t}{2} v_{ij} u_{ij}^n. \end{aligned}$$

Finally, grouping all terms evaluated at time level $n+1$ on the left-hand side we obtain,

$$\begin{aligned} & u_{ij}^{n+1} - r \left[u_{i+1,j}^{n+1} - 2u_{ij}^{n+1} + u_{i-1,j}^{n+1} \right] \\ & - r \left[u_{i,j+1}^{n+1} - 2u_{ij}^{n+1} + u_{i,j-1}^{n+1} \right] + \frac{i\Delta t}{2} v_{ij} u_{ij}^{n+1} \\ & = u_{ij}^n + r \left[u_{i+1,j}^n - 2u_{ij}^n + u_{i-1,j}^n \right] \\ & + r \left[u_{i,j+1}^n - 2u_{ij}^n + u_{i,j-1}^n \right] - \frac{i\Delta t}{2} v_{ij} u_{ij}^n. \end{aligned} \quad (14)$$

This relation defines, at each time step, a linear system for the unknown values u_{ij}^{n+1} on the interior grid points, subject to the Dirichlet boundary conditions imposed on the edges of the domain.

Sparse matrix formulation and numerical implementation

To advance the numerical solution in time using the Crank–Nicolson scheme derived in the previous subsection, Eq. (14) can be written in compact matrix form as

$$A \vec{u}^{n+1} = B \vec{u}^n, \quad (15)$$

where $\vec{u}^n$ collects all values u_{ij}^n at the internal grid points (i, j) at time step t_n , and A and B are sparse matrices encoding the finite-difference stencil and the potential.

We consider a square grid with M points in each spatial direction, including boundaries, so there are $M-2$ internal points along each axis. The total number of unknowns at each time step is therefore

$$N = (M-2)^2.$$

To store the two-dimensional field u_{ij}^n in a one-dimensional vector $\vec{u}^n$ we use a row-major ordering in which the x -index runs fastest:

$\vec{u}^n = (u_{1,1}^n \dots u_{M-2,1}^n, u_{1,2}^n \dots u_{M-2,2}^n \dots u_{1,M-2}^n \dots u_{M-2,M-2}^n)^T$. The corresponding C++ helper function `ij_to_k` implements the index map

$$k = (j-1)(M-2) + (i-1),$$

which takes a pair of internal indices (i, j) (with $i, j = 1, \dots, M-2$) and returns the associated zero-based vector index $k \in \{0, \dots, N-1\}$. This mapping is used systematically whenever matrix elements $A_{kk'}$ or $B_{kk'}$ are set.

In the implementation, the matrices A and B are represented using Armadillo's complex sparse matrix type `arma::sp_cx_mat`. A dedicated function `construct_AB_matrices` builds these matrices for a given grid size M , a complex scalar r , and two complex vectors $\vec{a}$ and $\vec{b}$ of length N , which provide the diagonal entries. For each internal grid point (i, j) the corresponding index k is computed via `ij_to_k`, and we set

$$A_{kk} = a_k, \quad B_{kk} = b_k.$$

The Crank–Nicolson discretisation of the Laplacian couples each point to its four nearest neighbours on the grid. Denoting by k_R and k_U the indices corresponding to the right $(i+1, j)$ and upper $(i, j+1)$ neighbours (when they exist), the off-diagonal elements are filled as

$$\begin{aligned} A_{kk_R} &= -r, & A_{k_R k} &= -r, \\ A_{kk_U} &= -r, & A_{k_U k} &= -r, \\ B_{kk_R} &= r, & B_{k_R k} &= r, \\ B_{kk_U} &= r, & B_{k_U k} &= r, \end{aligned}$$

which reproduces the five-point stencil structure implied by Eq. (14). The Dirichlet boundary conditions are already accounted for by restricting the construction to internal points only.

Before using these matrices in time evolution, we verified the correctness of the indexing and sparsity pattern. For this purpose, we chose values for the diagonals of matrices A and B according to expressions Eq. (16) and Eq. (17) and a constant non-zero complex value for r following Eq. (13).

$$a_k = 1 + 4r + \frac{i \Delta t}{2} v_{ij}, \quad (16)$$

$$b_k = 1 - 4r - \frac{i \Delta t}{2} v_{ij}. \quad (17)$$

Here, v_{ij} accounts for the elements of the matrix V representing the potential. For this verification process, we choose $V = \mathbb{I}$.

The matrices A and B were then assembled for different grid sizes (e.g. $M = 5$ and $M = 6$, corresponding to $N = 9$ Fig. 1 and $N = 16$ Fig. 2) and their structure was inspected using a helper function `print_sp_matrix_structure(const arma::sp_cx_mat&)`. The printed patterns display the expected block structure and the correct number of non-zero elements, confirming that the mapping $(i, j) \leftrightarrow k$ and the construction of A and B are consistent with the theoretical stencil.

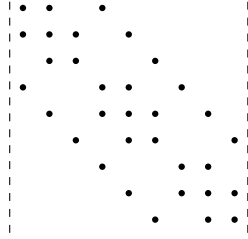


FIG. 1: Sparsity patterns for matrix of size 9×9 . Dots mean "non-zero value" and spaces mean "zero value".

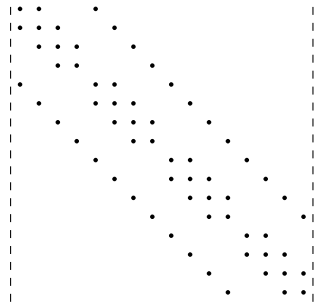


FIG. 2: Sparsity patterns for matrices of size 9×9 and 16×16 . Dots mean "non-zero value" and spaces mean "zero value".

Time stepping and solution of the linear system

Once the matrices A and B have been constructed, advancing the solution in a time loop means computing $\vec{u}^{n+1}$ from a given $\vec{u}^n$ by solving Eq. (15) at each step.

In the implementation this is done in two stages. First, the matrix multiplication

$$B \vec{u}^n = \vec{b} \quad (18)$$

is formed by a sparse matrix-vector multiplication. This operation is encapsulated in a small helper routine `matrix_mult`, which takes as input the sparse complex matrix B (stored as `arma::sp_cx_mat`) and the current solution vector $\vec{u}^n$ (stored as `arma::cx_vec`), and returns the complex vector $\vec{b}$.

In the second stage we solve the matrix equation

$$A \vec{u}^{n+1} = \vec{b} \quad (19)$$

for the unknown $\vec{u}^{n+1}$. The matrix A is sparse, complex and non-Hermitian, with a structure arising from the five-point stencil of the two-dimensional Laplacian and the local potential term. Rather than implementing a custom linear solver, we use the sparse direct solver provided by the Armadillo library. In particular, the routine `arma::spsolve` is employed with the "superlu" option, which delegates the factorisation and solution of Eq. (19) to the SuperLU library:

$$\vec{u}^{n+1} = A^{-1} \vec{b} \quad \leftarrow \quad \text{spsolve}(\text{u_next}, A, b, \text{"superlu"}).$$

This approach exploits the sparsity pattern of A and provides a robust and well-tested complex solver. In the time-stepping loop the two operations,

$$B \vec{u}^n = \vec{b}, \quad A \vec{u}^{n+1} = \vec{b},$$

are repeated for each time level until the final simulation time is reached, yielding the numerical evolution of the wave function $u(x, y, t)$.

Initial state and normalisation

The simulation is initialised with a two-dimensional Gaussian wave packet

$$u(x, y, t = 0) = \exp \left[-\frac{(x - x_c)^2}{2\sigma_x^2} - \frac{(y - y_c)^2}{2\sigma_y^2} + ip_x x + ip_y y \right], \quad (20)$$

where (x_c, y_c) denotes the centre of the packet, σ_x and σ_y are its initial widths in the x and y directions, and (p_x, p_y) are the corresponding momenta. The domain is $[0, 1] \times [0, 1]$ discretised with M grid points in each direction, so that $h = 1/(M - 1)$ and $x_i = ih$, $y_j = jh$ for $i, j = 0, \dots, M - 1$. Dirichlet boundary conditions are imposed by setting the wave function to zero on the outer grid points:

$$u(0, y, t) = u(1, y, t) = u(x, 0, t) = u(x, 1, t) = 0. \quad (21)$$

In the discrete setting this corresponds to fixing $u_{ij}^0 = 0$ for $i = 0$, $i = M - 1$, $j = 0$ or $j = M - 1$. The

remaining interior values u_{ij}^0 are initialised according to the Gaussian expression above. The resulting discrete wave function $U_0(i, j)$ is then normalised such that

$$\sum_{i,j} |u_{ij}^0|^2 = 1, \quad (22)$$

so that $|u_{ij}^0|^2$ can be interpreted as a probability associated with each grid cell.

The full procedure used to construct and normalise the initial state is summarised in Algorithm 1. In the implementation, the final normalisation loop over all grid points is parallelised using OpenMP, as this operation is embarrassingly parallel: each value $U_0(i, j)$ is scaled independently by the same normalisation factor.

Algorithm 1 Initialisation and normalisation

Input: $M, x_c, y_c, \sigma_x, \sigma_y, p_x, p_y$
 $h \leftarrow 1/(M-1)$
 Initialise complex matrix $U_0 \in \mathbb{C}^{M \times M}$ with zeros
 norm $\leftarrow 0$

for $j = 0, \dots, M-1$ **do**
 for $i = 0, \dots, M-1$ **do**
if $i = 0$ or $i = M-1$ or $j = 0$ or $j = M-1$ **then**
 $U_0(i, j) \leftarrow 0$ $\triangleright$ Dirichlet boundary conditions
 continue
 $x \leftarrow ih, \quad y \leftarrow jh$
 $\text{env} \leftarrow \exp\left(-\frac{(x-x_c)^2}{2\sigma_x^2} - \frac{(y-y_c)^2}{2\sigma_y^2}\right)$
 $\text{phase} \leftarrow \exp(i(p_x x + p_y y))$
 $U_0(i, j) \leftarrow \text{env} \times \text{phase}$
 norm $\leftarrow \text{norm} + |U_0(i, j)|^2$

$\alpha \leftarrow 1/\sqrt{\text{norm}}$
Parallel for all i, j (**normalisation loop**)
for $j = 0, \dots, M-1$ **do**
 for $i = 0, \dots, M-1$ **do**
 $U_0(i, j) \leftarrow \alpha U_0(i, j)$

Output: normalised initial state with $\sum_{i,j} |U_0(i, j)|^2 = 1$

Potential construction and double-slit geometry

The external potential $v(x, y)$ encodes the geometry of the box and the double-slit barriers. We work on the unit square $[0, 1] \times [0, 1]$, discretised with a uniform grid of M points in each direction. In the discrete representation, the potential is stored in a real matrix V with elements $v_{ij} = v(x_i, y_j)$, where $x_i = ih$ and $y_j = jh$. Except for the barrier regions, the potential is set to zero.

To construct the barriers we introduce a vertical wall of finite thickness in the x -direction, centred at a user-specified position x_{wall} (typically $x_{\text{wall}} = 0.5$). All grid

points whose x -coordinate satisfies

$$|x_i - x_{\text{wall}}| \leq \frac{d_{\text{wall}}}{2},$$

with d_{wall} the wall thickness, are considered to belong to the wall. For these grid columns we then define a set of N_s slits, each of vertical aperture a_{slit} , separated by wall segments of length L_{sep} . The slit pattern is constructed to be symmetric around the centre of the domain in the y -direction, $y_c = 0.5$. The total vertical span occupied by the slits and intermediate wall pieces is

$$L_{\text{tot}} = N_s a_{\text{slit}} + (N_s - 1) L_{\text{sep}},$$

and the lower edge of the first slit is placed at $y_c - L_{\text{tot}}/2$. For each slit $s = 0, \dots, N_s - 1$ we define an interval

$$\mathcal{S}_s = [y_s, y_s + a_{\text{slit}}], \quad y_s = y_c - \frac{L_{\text{tot}}}{2} + s(a_{\text{slit}} + L_{\text{sep}}),$$

and identify grid points (x_i, y_j) lying inside any of these intervals as belonging to a slit opening. The discrete potential is then set according to

$$v_{ij} = \begin{cases} v_0, & \text{if } (x_i, y_j) \text{ lies inside the wall but outside all slit } \mathcal{S}_s, \\ 0, & \text{otherwise,} \end{cases}$$

where v_0 is a large positive constant that represents an effectively impenetrable barrier. In this way the walls and the region between the slits are assigned a high potential, while the slits correspond to narrow low-potential channels through which the wave packet can propagate. Figure 3 shows an example of the resulting potential for a particular choice of parameters.



FIG. 3: Example of a double-slit potential $v(x, y)$ on an 80×80 grid with wall thickness $d_{\text{wall}} = 0.02$, wall position $x_{\text{wall}} = 0.5$, slit separation $L_{\text{sep}} = 0.05$, slit aperture $a_{\text{slit}} = 0.05$, and $N_s = 2$ slits. Dark regions represent the high-potential barrier; bright gaps correspond to the slit openings.

III. RESULTS AND DISCUSSION

Probability conservation for a free wave packet

As a first consistency check of the numerical implementation, we consider the evolution of a free Gaussian wave packet, i.e. with no double-slit barrier ($v_0 = 0$). The simulation parameters were chosen as: $h = 0.005$ (corresponding to $M = 201$ grid points in each direction), time step $\Delta t = 2.5 \times 10^{-5}$, total simulation time $T = 0.008$, initial packet centre $x_c = 0.25$, $y_c = 0.5$, widths $\sigma_x = \sigma_y = 0.05$, longitudinal momentum $p_x = 200$, and transverse momentum $p_y = 0$. In this configuration the wave packet propagates freely across the box, and the total probability

$$P^n = \sum_{i,j} |u_{ij}^n|^2 \quad (23)$$

should be conserved and remain equal to unity for all time steps n . In practice we monitor the deviation from unit probability, $|P^n - 1|$, as a function of time.

Figure 4 shows the deviation $|P^n - 1|$ when the total probability at step n is computed by an explicit sum over all elements of the state vector, i.e. by adding the squared moduli of the complex amplitudes one by one. In Fig. 5 we present the same quantity, but now evaluated using a compact vectorised Armadillo expression that directly accumulates the sum of $|u_k^n|^2$ over the internal degrees of freedom. In both cases we plot the deviation from unity rather than the probability itself, so that small numerical errors are visible on a logarithmic scale.

Both approaches give deviations of order 10^{-14} throughout the simulation, indicating that the Crank–Nicolson scheme together with the sparse linear solver conserves probability to within machine precision. The minor differences between the two curves arise from round-off effects and the slightly different ordering of floating-point operations, but they remain at the same numerical scale. This confirms that the time stepping and the matrix construction are internally consistent.

To test probability conservation in a more demanding situation, we next introduced the double-slit potential illustrated in Fig. 3. In this case the barrier height was set to $v_0 = 10^{10}$. All other simulation parameters were kept as in the previous case, except that the initial packet was broadened in the y -direction to $\sigma_y = 0.10$. The broader transverse profile and the presence of very steep potential barriers make this configuration a more stringent test of the numerical scheme.

As before, we monitored the deviation $|P^n - 1|$ of the total probability from unity at each time step. We again applied the two approaches explained above. For the first approach, the resulting deviation is shown in Fig. 6. The deviations remain at the level of $\times 10^{-14}$ and $\times 10^{-13}$ over the entire simulation, comparable to the free-packet case and again limited by floating-point round-off. The corresponding deviation for the second approach is displayed

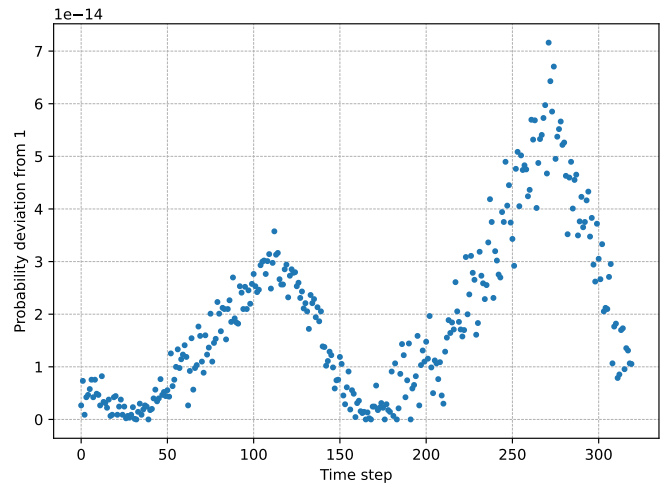


FIG. 4: Deviation of the total probability $|P^n - 1|$ as a function of time step index n for the free wave-packet simulation, obtained by explicitly summing the squared moduli of all components of the state vector at each time step.

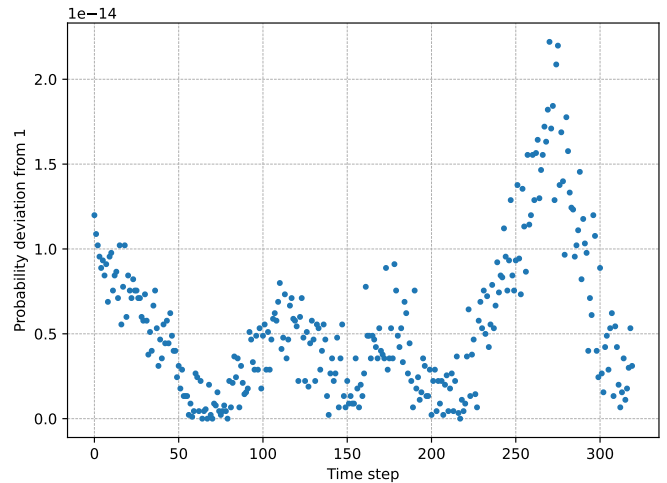


FIG. 5: Deviation of the total probability $|P^n - 1|$ as a function of time step index n for the free wave-packet simulation, computed using a vectorised Armadillo expression that directly evaluates the sum of $|u_k^n|^2$ over all internal grid points.

in Fig. 7. Here the deviations are slightly lower (10^{-14}). The small difference between the two curves reflects the different ordering of arithmetic operations rather than any physical effect, and both approaches confirm that the Crank–Nicolson scheme conserves probability to within numerical precision even in the presence of a double-slit potential and a broader initial wave packet.

Time evolution of the wave packet

We now turn to the qualitative behaviour of the wave function as it evolves in the double-slit potential illus-

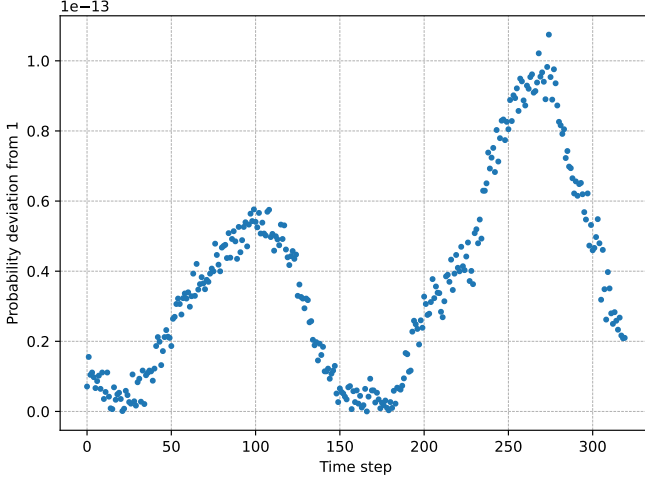


FIG. 6: Deviation of the total probability as a function of time step for the double-slit simulation with $v_0 = 10^{10}$ and $\sigma_y = 0.10$, computed by explicitly summing the squared moduli of all components of the state vector at each time step.

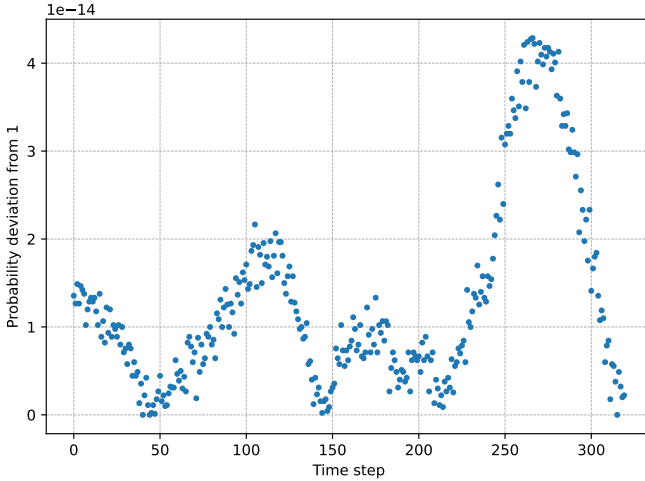


FIG. 7: Deviation of the total probability as a function of time step for double-slit simulation with $v_0 = 10^{10}$ and $\sigma_y = 0.10$, computed using a compact vectorised Armadillo expression that directly sums $|u_k^n|^2$ over all internal grid points.

trated in Fig. 3 with barrier height $v_0 = 1 \times 10^{10}$. The simulation parameters follow the specification of: spatial step $h = 0.005$ ($M = 201$), time step $\Delta t = 2.5 \times 10^{-5}$, total evolution time $T = 0.002$, and a Gaussian wave packet initially centred at $(x_c, y_c) = (0.25, 0.5)$ with widths $\sigma_x = 0.05$ and $\sigma_y = 0.20$. The longitudinal momentum is $p_x = 200$ and the transverse momentum is $p_y = 0$, so that the packet is initially directed straight towards the barrier. In the plots below we show snapshots at $t = 0$, $t = 0.001$ and $t = 0.002$, using independent colour scales in each panel so as to maximise the contrast at each time. In all cases the colour encodes the quantity indicated on the colour bar.

Figure 8 shows the time evolution of the probability field $p_{ij}^n = |u_{ij}^n|^2$. At $t = 0$ the probability is concentrated in a narrow Gaussian just to the left of the barrier (panel a). As the wave packet propagates and reaches the double slit, the probability becomes localised around the two openings (panel b), where significant flux is transmitted through the slits while the regions inside the wall remain essentially dark. At $t = 0.002$ we clearly observe the emergence of a double-slit interference pattern on the right-hand side of the barrier (panel c), with a series of probability maxima and minima arranged in a characteristic fringe pattern in the y -direction and symmetric about the midline $y = 0.5$. A simulation of the evolution of the probability field can be found in the GitHub repository².

To gain insight into the phase structure of the wave function, we also plot the real and imaginary parts of u_{ij}^n at the same time instants. The real part is shown in Fig. 9. Initially (panel a) the packet is a Gaussian envelope modulated by a plane wave in the x -direction, leading to vertical interference fringes across the packet. At $t = 0.001$ (panel b) these fringes become strongly distorted near the slits, where the wave splits into two transmitted components and undergoes diffraction. By $t = 0.002$ (panel c) the real part exhibits a rich, nearly circular fringe pattern on the right of the barrier, corresponding to the interference of the two outgoing wavefronts.

The corresponding plots for the imaginary part $\text{Im}(u_{ij}^n)$ are presented in Fig. 10. At $t = 0$ the imaginary component has the same Gaussian envelope as the real part but is phase shifted, resulting in an alternating fringe pattern in x (panel a). At intermediate times (panel b) the interaction with the barrier again produces strong local modulations near the slits. At $t = 0.002$ (panel c) the imaginary part shows the same global diffraction pattern as the real part, but with a relative phase shift between the fringes. Together, the real and imaginary colourmaps visualise the full complex interference pattern formed by the wave packet as it propagates through the double-slit geometry.

Detection probabilities on a distant screen

To further analyse the interference effects produced by the barrier, we consider the detection probability on a screen placed at $x = 0.8$, spanning the full vertical extent of the domain. Fixing the initial wave packet with parameters $(x_c, y_c) = (0.25, 0.5)$, $\sigma_x = 0.05$, $\sigma_y = 0.20$, $p_x = 200$ and $p_y = 0$, and evolve the system to time $t = 0.002$. At this final time we extract the one-

² https://github.uio.no/teresm/Comp.Phys.Project/blob/master/P5/plots/probability_evolution.gif

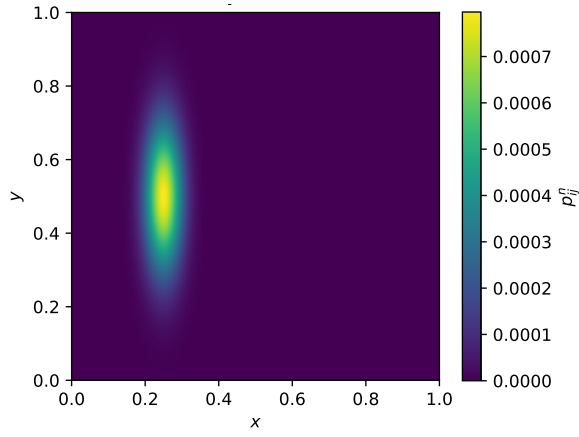
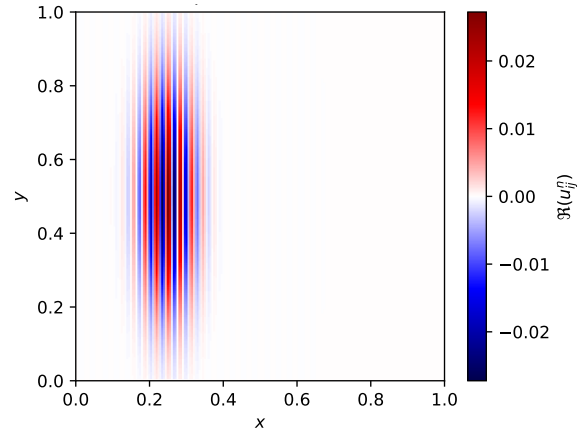
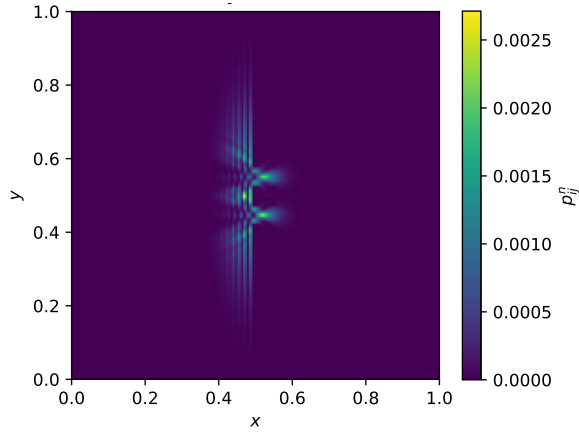
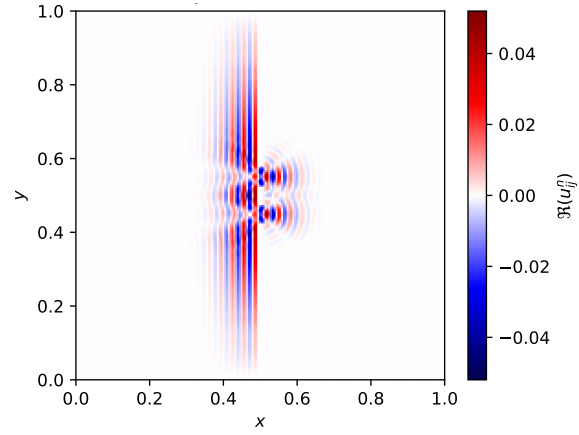
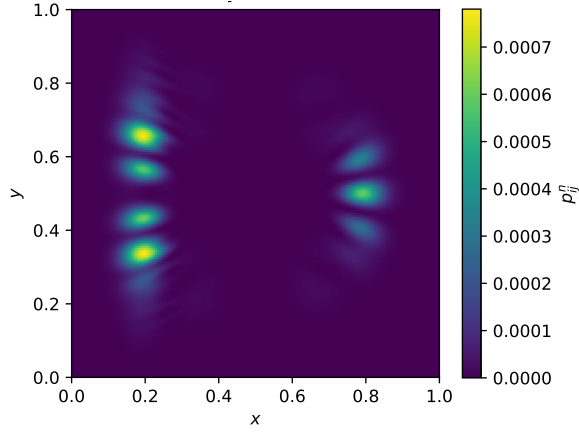
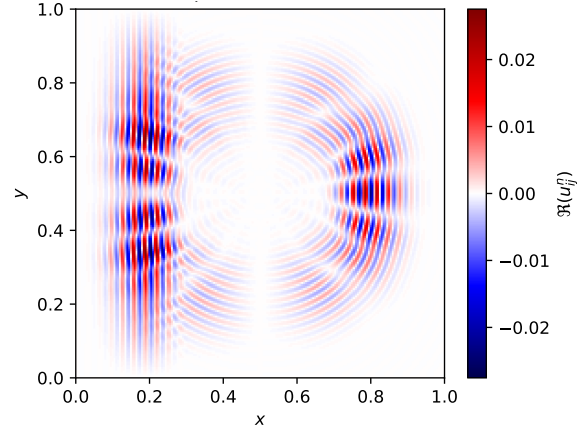
(a) $t = 0$ (a) $t = 0$ (b) $t = 0.001$ (b) $t = 0.001$ (c) $t = 0.002$ (c) $t = 0.002$

FIG. 8: Snapshots of the probability field $p_{ij}^n = |u_{ij}^n|^2$ at three different times. Each panel uses its own colour scale, normalised to the maximum value of p_{ij}^n at that time.

FIG. 9: Real part of the wave function, $\text{Re}(u_{ij}^n)$, at the same three times as in Fig. 8. The colour scales are symmetric about zero and adapted to the maximum absolute value of $\text{Re}(u_{ij}^n)$ in each panel.

dimensional probability distribution along the screen,

$$p(y \mid x = 0.8, t = 0.002) \propto |u(x = 0.8, y, t = 0.002)|^2, \quad (24)$$

and normalise it such that $\sum_j p(y_j \mid x = 0.8, t = 0.002) = 1$. We repeat this procedure for three differ-

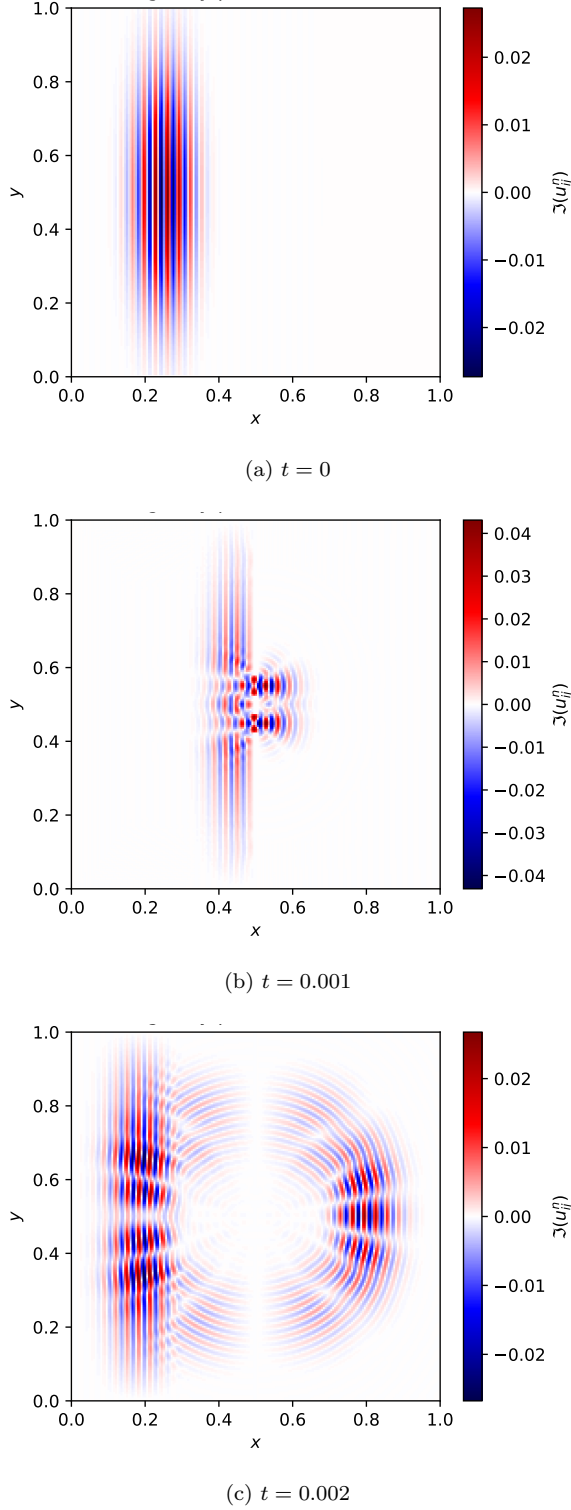


FIG. 10: Imaginary part of the wave function, $\text{Im}(u_{ij}^n)$, at the same times as in Fig. 8. As for the real part, the colour scales are symmetric about zero and adjusted independently for each panel.

ent barrier configurations: a single slit, a double slit, and a triple slit. In all cases the wall parameters are $v_0 = 1 \times 10^{10}$, wall thickness $d_{\text{wall}} = 0.02$, wall position

$x_{\text{wall}} = 0.5$, slit separation $L_{\text{sep}} = 0.05$, and slit aperture $a_{\text{slit}} = 0.05$, while only the number of slits is changed.

Figure 11 shows the resulting detection probability for the single-slit configuration. The distribution exhibits a single dominant maximum near the central region of the screen, with smaller side lobes arising from diffraction at the slit edges.

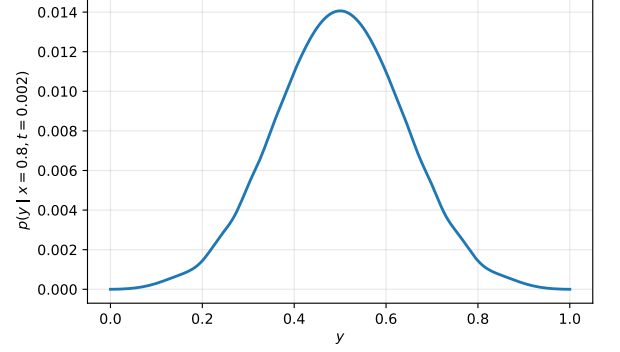


FIG. 11: Normalised detection probability $p(y | x = 0.8, t = 0.002)$ on the screen for the single-slit potential.

For the double-slit potential, shown in Fig. 12, the screen probability develops a pronounced interference pattern. Instead of a single peak, we observe a series of alternating maxima and minima along the y -axis, with a central maximum around $y = 0.5$. The fringe spacing is set by the slit separation L_{sep} and the effective wavelength of the incident packet.

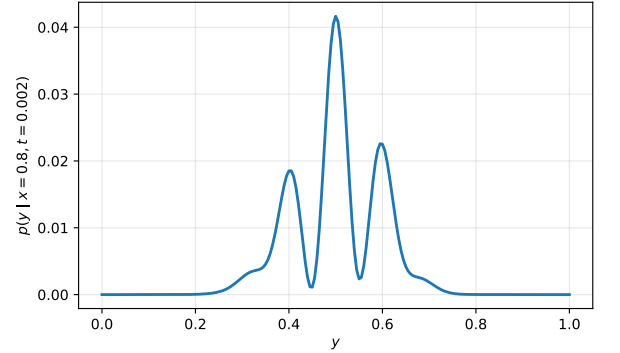


FIG. 12: Normalised detection probability $p(y | x = 0.8, t = 0.002)$ on the screen for the double-slit potential.

Finally, Fig. 13 displays the result for a three-slit barrier. The pattern becomes more intricate: two principal peaks appear instead of one. The structure can be interpreted as the coherent superposition of three diffracted wavefronts, where interference between each pair of slits contributes to the final pattern. Compared to the double-slit case, the increased number of slits sharpens the peaks and enhances contrast in specific regions of the screen, while still respecting the overall normalisation of the detection probability.

Together, these three cases illustrate how the spatial structure of the potential directly shapes the detection

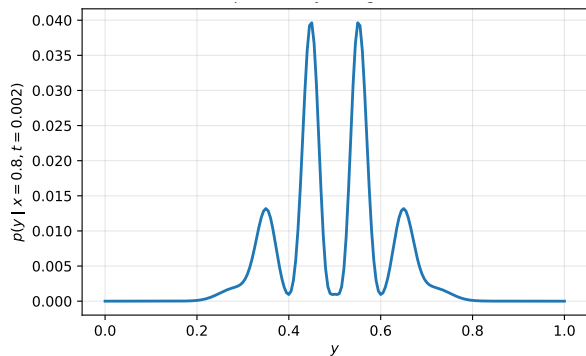


FIG. 13: Normalised detection probability $p(y | x = 0.8, t = 0.002)$ on the screen for the three-slit potential.

probabilities: increasing the number of slits transforms a simple diffraction pattern into a rich interference pattern, in agreement with basic wave optics and quantum mechanics expectations.

IV. CONCLUSION

In this work we have implemented and tested a Crank–Nicolson solver for the two-dimensional time-dependent Schrödinger equation on a finite box with Dirichlet boundary conditions. The discretisation leads to a sparse complex linear system at each time step, which we formulated in matrix form using a five-point stencil and solved with a sparse direct method from the Armadillo library.

Probability conservation tests for a freely propagating Gaussian wave packet demonstrated that the scheme preserves the total norm of the wave function to within machine precision over hundreds of time steps, both in the absence and in the presence of a steep double-slit potential. This confirms the internal consistency and numerical stability of the implementation.

Building on this foundation, we used the solver to study the evolution of a Gaussian wave packet incident on one-, two-, and three-slit barriers. For the double-slit potential we visualised the full time development of the probability density and of the real and imaginary parts of the wave function, and we observed the emergence of a clear interference pattern behind the barrier. By extracting normalised detection probabilities on a screen placed at $x = 0.8$, we quantified how the interference fringes change as the number of slits increases, finding the expected progression from a broad single-peaked diffraction pattern to increasingly structured multi-peak interference profiles.

The code developed for this project provides a flexible framework for exploring more general quantum scattering problems. Possible extensions include studying different initial states or potential geometries, incorporating absorbing boundary layers to reduce reflections at the box edges, and investigating higher-order time integrators or alternative sparse solvers. Nevertheless, the present results already show that a relatively simple Crank–Nicolson discretisation, combined with modern sparse linear algebra tools, is sufficient to capture the essential physics of two-dimensional quantum interference in the iconic double-slit setup.

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 - [3] Matplotlib: Visualization with Python. <https://matplotlib.org/stable/>, 2025. Accessed: 2025-12-05.
 - [4] Anders Kvellestad. Lecture notes, 2025.